

Vedant Deshmukh

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EXPERIENCE

Nimbb! (payment orchestration)

Software Development Engineer - I

Software Development Intern

Mumbai, MH

February 2025 – Present

October 2024 – February 2025

- Built a **dynamic payment routing engine** replacing static priority-based routing across **~30 PSP integrations**, computing per-partner success-rate scores over rolling windows sliced by payment mode, sub-payment mode, and configurable dimensions (bank, network, issuer, BIN); simulation on one month of historical transaction data showed a **~7% improvement in transaction success rate**.
- Designed two routing strategies, **pure success-rate** and **volume-aware exploration** (sampling traffic to low-volume PSPs for reliable scores), with cooldown-based thrashing prevention and automatic preferred-PSP recovery.
- Redesigned master data caching with a centralized declarative framework, removing async I/O overhead from coroutine-based cache reads and standardizing cache behavior across services, cutting **memory usage by ~32%** and **CPU by 0.2 cores per pod** (~1.6 cores across 8 API pods).
- Implemented **PCI DSS-compliant** payment flows (encryption, least-privilege access, audit logging) and modernized Kotlin services to **JDK 17** with hardened PSP error handling.
- Extended and maintain **Skypay UOP**, a SOAP↔REST proxy carrying **Air India Express's entire payment traffic** through Nimbb!; shipped new payment flows and requirement-driven changes for legacy airline systems.

Graduate Researcher & Co-Author

Mumbai University (in collaboration with Fortis Hospital)

August 2024 – February 2025

Mumbai, MH

- Co-authored a research paper awarded **Best Application Paper** at an IIT Roorkee-hosted conference (in press, **Springer Proceedings in Mathematics & Statistics**, 2026), on diagnostic accuracy and explainability in medical imaging AI.
- Designed a **CNN-based dual-task mammography system** (customized Xception) for breast-density (4-class) and malignancy (binary) classification on a **7,632-image** clinical dataset, iterating from a 75% baseline to **99.7% published density-classification accuracy**.
- **Invited speaker** at Xavier Institute of Engineering: delivered sessions on programming with biomedical imaging and NLP in healthcare (Sept 2025).

PROJECTS

Command Centre | Kotlin, Spring Boot, AWS, PostgreSQL, Hazelcast, Docker

August 2025 – Present

- **Leading development** of an internal payments operations platform replacing a legacy ERP, with **multi-tiered RBAC** and **bi-directional ERP synchronization** for parallel workflows during migration.
- Designed and built **Orbit**, the entity-level configuration architecture managing PSP credentials, payment modes, and fee plans across a **Partner→Merchant→Sub-merchant hierarchy**, with config versioning and effective-config resolution across levels.
- Built the data and validation layer of **Pulse**, the routing rules-authoring backend, replacing brittle per-row routing configs with **reusable rule blocks**; added revision-based editing with **pessimistic locking** against concurrent edits, cross-entity validations, and bulk CSV import.
- Defined the platform's change-tracking architecture: a generic Hibernate event listener feeding an external change-streamer via tracker tables, with **AOP-based audit diffing** for externally-written entities; documented the decision framework adopted across teams.
- Optimized bulk exports of up to **500,000 orders** via batching and streaming, eliminating OOM failures.

Partner Success Platform | Python, asyncio, SQLAlchemy

November 2024 – January 2025

- Architected an async internal platform for partner-success scripts and operational workflows, with separate read/write DB pools for throughput isolation; **led a 4-5 person partner-success team** through framework onboarding, code reviews, and task breakdown.
- Refactored the legacy codebase from **25,000 to 7,000 LOC**, dramatically improving maintainability and onboarding speed.

EDUCATION

Thadomal Shahani Engineering College

Bachelor of Engineering in Computer Engineering

Mumbai, MH

August 2020 – July 2024

TECHNICAL SKILLS

Languages: Java, Kotlin, Python, SQL, JavaScript, Bash

Backend: Spring Boot, Spring Batch, Ktor, Tornado (asyncio), Flask, Hibernate/JPA, REST APIs, SOAP

Data & Infra / ML: PostgreSQL, Hazelcast, AWS, Docker, Git; CNNs, transfer learning (Xception), TensorFlow/Keras

Domain: Payment orchestration, PSP integrations, PCI DSS compliance, RBAC, audit systems